Parth Mishra

AI Engineer | Python Backend & LLM Applications
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SUMMARY

Final-year B.Tech CSE (AI/ML) student who builds LLM apps and Python backends. Wrote a vector database from scratch (HNSW, 38x faster than brute force) with a local RAG pipeline. Also shipped a streaming Groq recruiter assistant with rate limits, and a Claude meal tracker that returns structured JSON. Looking for AI Engineer and SDE Intern roles.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL
LLM & AI: Claude API (Anthropic), Groq API, Ollama, Prompt Engineering, Structured JSON Outputs, RAG Pipelines, Embeddings (nomic-embed-text), Vector Search (HNSW, KD-Tree), Streaming Inference, NumPy, Scikit-Learn
Backend & APIs: FastAPI, Flask, Pydantic, Node.js, Express.js, REST APIs, Asynchronous Programming, Rate Limiting (slowapi), JWT Authentication
Data & DevOps: PostgreSQL, MySQL, Docker, Git, GitHub, Linux/Unix Shell, Render, Vercel, Pytest
Frontend: React, Vite, React Native, HTML5, CSS3
In Progress: LangGraph, Model Context Protocol (MCP), FAISS/ChromaDB benchmarking, LLM-as-a-Judge eval suite

PROJECTS

VectorDB: Vector Database & RAG Pipeline | Python, Flask, NumPy, Ollama | GitHub
- Cut nearest-neighbor search to sub-15ms over 50,000+ 768-dim embeddings, 38x faster than brute force. Built HNSW and KD-Tree indexes from scratch and benchmarked precision and latency against a brute-force baseline.
- Built the full local RAG loop: nomic-embed-text embeddings, HNSW retrieval, answers from an Ollama LLM, all behind Flask REST endpoints (/insert, /search, /benchmark). A PCA dashboard plots the semantic clusters in 2D.

AI Portfolio & Recruiter Assistant | React, Vite, FastAPI, Groq API | Live | GitHub
- Streamed answers to recruiter questions in real time through FastAPI and the Groq API (openai/gpt-oss-20b). The prompt is grounded on resume and project data, so the bot sticks to facts.
- Capped abuse with slowapi rate limits: 20 req/min on chat, 5 req/min on the JD matcher. The JD evaluator returns a suitability score, missing skills, and interview tips. Frontend on Vercel, backend on Render.

Eve Diagnostic Booking Service | FastAPI (Python 3.12), PostgreSQL 16, Pydantic v2, Docker Compose, Pytest | GitHub
- Stopped double-booking and duplicate charges under concurrent requests and webhook retries, using pessimistic row locking (SELECT ... FOR UPDATE) and an idempotency-key ledger with a transactional outbox. Async Pytest suites reproduce the conflicts; Docker Compose runs the API and database.

AI Meal Tracker | React, Vite, Claude API (claude-sonnet-4-5), Prompt Engineering | GitHub
- Turned free-text Indian vegetarian meal descriptions into structured JSON macro breakdowns. Prompts encode home-cooking portion sizes, which avoids generic USDA-style estimates. No backend: the browser calls the API directly.

Autonomous Job-Hunt Pipeline | Python 3.12, IMAP/SMTP, XeTeX | GitHub
- Dropped 95%+ of irrelevant postings before any LLM call, using regex filters on title, batch, and location. Survivors get LLM screening, a tailored XeTeX resume, and an emailed application, all logged in a CSV ledger.

EXPERIENCE

Software Engineering Intern (Front-End & Systems) | Craftedge Academy | Remote / Bhopal | Oct 2025 - Apr 2026
- Cut initial page load latency 28% on client-facing platforms with 1,000+ active learners. The fixes: React code-splitting, lazy-loading, and asset minification. Mobile layout shifts are gone too.
- Reduced redundant network requests 35%, with zero regressions across releases, by adding centralized client-side state and debounced REST queries.
- Shipped modular, responsive React components that passed 100% of QA verification.

EDUCATION

Vellore Institute of Technology (VIT Bhopal) | B.Tech, Computer Science and Engineering (AI/ML) | CGPA: 8.67 | Aug 2023 - May 2027
Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

- **CS50's Introduction to Artificial Intelligence with Python**, Harvard University (online) | Oct 2026 | Verify: 3e7e245a
- **Oracle Certified Foundations Associate, Agentic AI**, Oracle University | Sept 2026 | Verify: 103546830AAI26OFA
- **Applied Machine Learning in Python**, University of Michigan (Coursera) | Jan 2026 | Verify: G1Q6ZYPKPWP4

ACHIEVEMENTS

- Ranked top 3 in the district among 1,200+ students in a Mathematics Olympiad.
- Ranked top 5 among 7,000+ students in a Mathematics Olympiad.
- Solved 250+ algorithmic problems on LeetCode, GeeksforGeeks, and HackerRank (DP, graphs, trees, two pointers). Solutions: GitHub.